



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

BDCRS401

REINFORCEMENTS STEEL BARS WORKS

Perform Reinforcements steel bars works

Competence

RQF Level: 4

Learning Hours



80

Credits: 8

Sector: Construction and Building Services

Trade: Building Construction

Module Type: Specific

Curriculum: CBSBDC4001- TVET Certificate 4 in Building Construction

Copyright: © Rwanda TVET Board, 2023

Issue Date: April ,2023

Purpose statement	This module describes the knowledge, skills, and attitude to be acquired to perform reinforced steel bars works as well as performing all steps involved in making steel bars framework on the site. It is intended to be pursued by the learner at TVET level IV in building construction. At the end of this module, the learner will be able to select tools, equipment and materials to be used for steel bar works, interpret steel bars drawings, calculate cutting length, cut steel bars, bend steel bars and tie steel bars.					
Leaning assumed to be in place	Building construction drawing, Concrete works, Mathematics, Physics and Safety at work place.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		25%	Formative assessment	30%	50%
	Practical work:		75%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Perform preliminary works	1.1. Construction drawing is correctly interpreted based on information required.
	1.2. Materials are properly prepared according to the work to be done.
	1.3. Tools are appropriately prepared according to the work to be done.
	1.4. Equipment are appropriately prepared in accordance with their function.
	1.5. Workplace is properly prepared according to the work to be done.
2. Make steel bar frames	2.1. Bending table is accurately set according to the rebar size to be bent
	2.2. Steel bars are accurately measured according to the place to be used.
	2.3. Steel bars are correctly cut according to the required length.
	2.4. Steel bars are properly bent according to bar bending schedule (BBS).
	2.5. Steel bars are tightly tied according to the structure member shape.
3. Adjust steel bar frames in the	3.1. Steel bar frames are properly installed in the formwork according to concrete cover required and with respect of overlap standards.

formwork	3.2. Steel bar frames are correctly aligned and levelled according to the structure element.
	3.3. Steel bar frames are adequately fixed according to the structure element.
	3.4. Concrete spacers are properly placed according to the designed concrete cover.
4. Clean tools, equipment and the workplace	4.1. Working area is properly cleaned after work.
	4.2. Used tools and equipment are properly cleaned according to their type.
	4.3. Used tools and equipment are correctly stored according to their category and status.

Intended Knowledge, Skills and Attitude

Knowledge	Skills	Attitude
✓ Identify steel bar works materials ✓ Identify different building elements ✓ Identify tools and equipment used for steel bars works	✓ Handle and use Tools, Equipment and material ✓ Read and interpret bar bending schedule and engineering plans ✓ Plan work and procedures of activities ✓ Apply Measurement skills ✓ Mark out work areas ✓ Use both hand and power tools ✓ Cut and bend stirrups and hooks ✓ Operate bar cutting and bending machines ✓ Lay out steel reinforcing bars, rods, mesh and formwork ✓ Fit spacers and supports ✓ Join steel with wires or by welding ✓ Fix steel to solid bases ✓ Fix and set steel beams	✓ Demonstrate awareness of workplace hazards and associated control measures ✓ Demonstrate persistent, team working, flexibility, punctuality at work ✓ Demonstrate the ability to Work independently with minimal supervision ✓ Respect time assigned to every task ✓ Being stress tolerant and self-motivated

	✓ Prepare technical report of the work done	
--	---	--

Course content

Learning outcomes	At the end of the module the learner will be able to: 1.Perform preliminary works 2.Make steel bar frames 3.Adjust steel bar frames in the formwork 4.Clean tools, equipment and the workplace
Learning outcome 1: Perform preliminary works	Learning hours: 15

Indicative content

- **Selection and use PPE, tools, materials and equipment for reinforcement steel bars works**
 - Classification and function of PPEs
 - ✓ Head protection
 - ✓ Body protection
 - ✓ Hearing protection
 - ✓ Eyes and face protection
 - ✓ Foot protection
 - ✓ Hand protection
 - ✓ Respiratory protection
 - Classification of tools and equipment
 - ✓ Straightening Tools and equipment
 - ✓ Measuring tools
 - ✓ Cutting Tools and equipment
 - ✓ Bending Tools and equipment
 - Identification of materials used in reinforcements steel bars
 - ✓ Steel bars
 - ✓ Wires
 - ✓ spacers
 - Identification of steel bars
 - ✓ Mild steel bars
 - ✓ Deformed steel bars
 - Identification of wires
 - ✓ Galvanized
 - ✓ Non galvanized
- **Cleaning workplace**
 - Methods of cleaning work place
 - ✓ Removal of wastes

➤ Provision of safeguard and signs • Interpretation of drawing and Technical Specifications ➤ Dimensioning, lettering, scale and symbols ✓ Bar codes ✓ Common symbols ✓ Reinforcement detailing for structural elements	
Resources required for the learning outcome	
Equipment	Electric bar cutter machine, Electric bar bending machine, Bending table, PPEs
Materials	Hack saw blade, cutting disc, Binding wires, Rebars, Building line
Tools	Hack saw, Hammers, Grinder machine, Tie wire & reels, Hand twister, Manual steel cutter machine, Plumb bob, building square, Measuring tape, Spirit level, Hand saw
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • Brainstorming
Formative assessment methods	• Matching questions • Multiple choice • Short questions • Open ended questions

Learning outcome 2: Make steel bar frames	Learning hours: 35
Indicative content	
• Introduction to BBS (bar bending schedule) ➤ Importance of BBS • Calculation of cutting length and number of bars based on dimensions of structural element ➤ Calculating the Cutting length of stirrups ✓ Rectangular ✓ Square ✓ Helical ✓ Circular ✓ Triangular ✓ diamond ➤ Calculation of cutting length and number of reinforcement steel bars ✓ footing ✓ column ✓ beam ✓ slabs ✓ stairs ✓ crank bar ✓ chair bar	

- **Marking and measuring of reinforcement steel bars**
 - Measurement of steel bars according to the specifications of drawings
 - Marking reinforcement steel
- **Cutting of steel bars according to the drawings or the cutting length calculated**
 - Equipments required for cutting reinforcement steel bars
 - cutting reinforcement steel bars according to the requirements
- **Bending reinforcement steel bars**
 - Methods of bending reinforcement steel bars
 - ✓ Hand method
 - ✓ Machine method
 - Setting of Bending table
 - ✓ Set and fix pins based on bar size
 - ✓ Set and fix pins based on size of stirrups
 - Bending basic components of steel works.
 - ✓ Standard hooks
 - ✓ Stools
 - ✓ Chairs
 - ✓ Cranked reinforcement steel bars
 - ✓ stirrups according to their types
- **Tying reinforcement steel bars for structural elements**
 - Identification of tying rebars technics
 - ✓ Simple tie or snap tie
 - ✓ Snap tie with a round turn or wall tie
 - ✓ Double-strand simple tie
 - ✓ Saddle or U tie
 - ✓ Saddle or U tie with a twist
 - ✓ cross figure eight tie
 - Application of tying Technics
 - ✓ Tying of main steel bars to stirrups
 - ✓ Fixing of support bars
 - ✓ Tying of distribution reinforcements

Resources required for the indicative content

Equipment	Electric bar cutter machine, Electric bar bending machine, Bending table, PPEs
Materials	Hack saw blade, cutting disc, Binding wires, Rebars, Building line
Tools	Hack saw, Hammers, Grinder machine, Tie wire & reels, Hand twister, Manual steel cutter machine, Plumb bob, building square, Measuring tape, Spirit level, Hand saw
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • brainstorming
Formative assessment methods	• Matching questions • Multiple choice

- Short questions
- Open ended questions
- Practical exercises

Learning outcome 3: Adjust steel bar frames in the formwork

Learning hours: 20

Indicative content

- **Assessment of existing formworks**
 - Types and materials of formworks
 - Timber formwork
 - Steel formwork
 - PVC formwork
 - Assessing the Requirements for structural elements formworks
 - Verticality
 - Horizontality
 - Checking accuracy of measurements
- **Transport steel frame and Install steel frame**
 - Transportation of reinforcement steel bar frame
 - Overlapping rules
 - Overlapping reinforcement steel bars
- **Alignment and levelling frame according to the structure element**
 - Check Alignment, Concrete Cover and Levels
 - Horizontal and vertical alignments
 - Cover and supports types
 - Importance of concrete cover
 - Disadvantages of excess and insufficient cover
- **Fixing reinforcement steel frame**
 - Placing of reinforcement steel frame
 - Placing of spacers according to designed concrete cover

Resources required for the indicative content

Equipment	Electric bar cutter machine, Electric bar bending machine, Bending table, PPEs
Materials	Cutting disc, Binding wires, Rebars, spacers, building line
Tools	Pliers, Hack saw, Hammers, Grinder machine, Tie wire & reels, Hand twister, Manual steel cutter machine, Plumb bob, building square, Measuring tape, Spirit level, Hand saw
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • Brainstorming

Formative assessment methods	• Matching questions • Multiple choice • Short questions • Open ended questions • Practical exercises
------------------------------	---

Learning outcome 4: Clean tools, equipments and the workplace	Learning hours:10
Indicative content	
• Cleaning of workplace after work ➤ Removal of waste materials ➤ Materials for cleaning workplace ➤ Methods of cleaning workplace ✓ By Water ✓ By Brush ✓ By Air compressor • Cleaning tools and equipments ➤ Methods of cleaning tools and equipments ✓ By Water ✓ By Brush ✓ By oil ✓ By air compressor ➤ Purposes of cleaning tools and equipments • Storing tools and equipments ➤ Selection of storing tools and equipments ➤ Ways of storing tools and equipments ➤ Points to follow in storing tools and equipments ✓ Select area ✓ Prioritize tools ✓ Separate tools ✓ Discard unused tools ✓ Label the place based on kinds of tools and equipments ➤ Advantages of storing tools and equipments	

Resources required for the indicative content	
Equipment	Air compressor, PPE
Materials	Water, soap, oil
Tools	Brooms, brush, buckets, floor scrubbing, vacuum cleaner
Facilitation techniques	• Group discussion • Trainer guided • Demonstration • Brainstorming
Formative assessment methods	• Matching questions • Multiple choice • Short questions • Open ended questions • Practical exercises

Integrated/Summative assessment (For specific module)

Integrated situation

Muhanga District has launched the master plan of Muhanga city to be implemented from 2020. From this, Commercial building at the city center must be multistory buildings (R2,3&4). Murenzi as the owner of one commercial building located at Muhanga district, Nyamabuye sector, offer a tender of rehabilitation of his building.

ACEB Ltd won this tender and reinforced cement concrete is needed to be used for structural members, hence there is a need of steel fixer for making reinforcement steel bar framework and fix the reinforcement steel bars framework in formwork at the construction site. The formwork makers are available at site.

As a steel fixer from ACEB Ltd, you are asked to do above tasks for one footing, one column.

Instructions:

- Footing is in square form of 120cm and the height of 40cm.
 - Bottom reinforcements 16mm@200mm
 - Top reinforcements 12mm@200mm
 - Hook=9d
 - Cover=75mm
- Height of column is 3m, it is square form of 20cm
 - Diameter of stirrup is 8mm @160mm
 - Main bars of 4Y12

- Hook= 9d
- Cover=40mm.
- The work should be done in 4 hours for each structural member.

Resources

Tools	Hammer, Tape measure, Vise, Pliers, Pincers, Hacksaw frame, Bender bar, Vernier caliper, straightening bar
Equipment	Cutting machine, bending machine, Bending table, PPEs
Materials/ Consumables	Steel bars, Concrete Spacer, Binding Wires, Cutting disc, Hacksaw blade

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: Perform preliminary works (15%)	1.1. Appropriate tools are selected according to the work to be performed	Tools are well selected (Hammer, Tape measure, Vise, Pliers, Bender bar, Vernier caliper, straightening bar)			3
	1.2. Equipment is selected according to the work to be performed	Equipment are properly selected (Cutting machine, bending machine, Bending table, PPEs)			3
	1.3. Adequate materials are selected according to the standard	Materials are properly selected (Steel bars, Concrete Spacer, Binding Wires, Cutting disc Hacksaw blade)			4
	1.4. proper preparation of work place	Workplace is properly prepared			5
Learning outcome 2: Make steel bar frame (40%)	2.1. accurate setting of bending table according to rebar size to bend	Pins are accurately set according to frame dimensions			3
		Pins are accurately set according to bar sizes			3
		Bending table is accurately set			4
	2.2. accurate measuring of steel bars	Steel bars are accurately measured			5
	2.3. corrective cutting of steel bars according to the required length.	Steel bars are accurately cut			5
	2.4. proper bending of steel bars according to bar bending schedule	Hook dimensions are respected			3
		Bending angles are respected			3
		Steel bars are properly bent			4
	2.5. tightly tying of steel bars	Steel bars are tightly tied			3

	according to structural members shape	Bars size is respected			3
		Bars spacing is respected			4
Learning outcome 3: Adjust steel bars frames in formwork (34%)	3.1. Proper installation of steel bars frames according to concrete cover required with respect of overlapping required	Steel bars frames are well transported			5
		Steel bars frames are well fixed			5
		Concrete cover is respected			5
		Overlapping standards are well respected			5
	3.2. correct alignment and levelling of steel bars frames.	Steel bars frames are well levelled			5
	3.3. adequate fixing steel bars frame according to structural elements.	Steel bars frame are adequately fixed			5
	3.4. proper placing of concrete spacers according to designed concrete cover	Concrete spacers are properly placed			4
Learning outcome 4: Cleaning tools, equipment and workplace (11%)	4.1. proper cleaning of tools and equipment	Used tools are properly cleaned			2
		Equipment are well cleaned			2
	4.2. proper storing of tools and equipment according to their kind	Tools and equipment are properly stored			3
	4.3. proper cleaning of workplace	Workplace is well cleaned			2
	4.3. time management	The work is done in 8 hours			2
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. <https://www.studocu.com/cl/document/universidad-de-chile/hormigon-estructural/div-syd-detailing-of-reinforcement-in-concrete-structures/28741127>
2. <https://www.maine.gov/mdot/bdg/docs/bpdg/chpt8.pdf>
3. <http://www.diynetwork.com/how-to/rooms-and-spaces/floors/how-to-install-a-hardwood-floor>
4. <https://www.csdcindia.org/wp-content/uploads/2018/12/Bar-Bender-and-Steel-Fixer-Level-4.pdf>
5. https://www.google.rw/search?dcr=1&source=hp&ei=n68CWqOfC8PxUo68pKgG&q=wood+floor+in+stallation&oq=wood+floor+inst&gs_l=psyab.1.0.0l10.1354108.1366976.0.1370892.43.24.16.0.0.0.345.315.1.2-1j9.11.0....0...1.1.64. psyab..17.26.3333.6..35i39k1j0i10k1j0i22i30k1.397.a
6. Brian F. P., Stagg W. D. (2007). Concrete technology: An Encyclopedia (4th edition), Wiley Blackwell
7. Taylor J.B., (1990). Concrete (5 th Edition), Longman
8. William., George B. (2013). Plastering Plain and Decorative: (4th Revised Edition), Taylor & Francis Ltd